

RAVEN AI Uncovers 118 New Planets, Including Ultra-Short Orbits and Worlds in the 'Neptunian Desert'

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Sifting through data from millions of stars to spot a planet is like finding a needle in a cosmic haystack. Now, a new artificial intelligence system called RAVEN has done just that, confirming 118 previously unknown **exoplanets** and flagging over 2,000 high-quality candidates from NASA's **Transiting Exoplanet Survey Satellite (TESS)**. Developed by astronomers at the University of Warwick, RAVEN (Rapid Automated Validation of **Exoplanets** from Noise) was trained on hundreds of thousands of simulated planets and false signals, such as **eclipsing binary stars**. This allows it to distinguish genuine planetary **transits**—the tiny dips in starlight when a planet crosses its host star—from impostors with unprecedented accuracy. "Using our newly developed RAVEN **pipeline**, we were able to validate 118 new planets, and over 2,000 high-quality planet candidates, nearly 1,000 of them entirely new," said Dr. Marina Lafarga Magro, lead author of the study published in *MNRAS*. The team focused on short-period planets—those completing an orbit in less than 16 days—using observations from over 2.2 million stars collected during TESS's first four years. Among the newly confirmed worlds are several extreme types. Some are **ultra-short-period planets** that whip around their stars in under 24 hours. Others reside in the so-called "**Neptunian desert**," a region where few Neptune-sized planets are expected to exist, challenging current theories of planetary formation. The survey also revealed tightly packed multi-planet systems, including previously unknown pairs orbiting the same star. "The challenge lies in identifying if the dimming is indeed caused by a planet or by something else," explained Dr. Andreas Hadjigeorgiou, who led RAVEN's development. "Its strength stems from our carefully created dataset of realistically simulated planets and other astrophysical events." Beyond individual discoveries, the team used this validated sample to measure how common close-in planets are around Sun-like stars. In a companion study, they found that about 9-10% of such stars host a close-in planet, a result that aligns with earlier Kepler mission findings but with uncertainties reduced by up to a factor of ten. "For the first time, we can put a precise number on just how empty this 'desert' is," said Dr. Kaiming Cui, lead author of the population study. Dr. David Armstrong, senior co-author, emphasized that RAVEN's reliability makes it more than a list of candidates—it's a robust tool for mapping planetary populations. The team has released interactive catalogs for other scientists to explore, marking a new era where AI accelerates the discovery of alien worlds while sharpening our understanding of the cosmos.

Word Treasure

exoplanet -- A planet that orbits a star outside our own solar system.

transit -- The slight dimming of a star's light when a planet passes in front of it, used to detect exoplanets.

eclipsing binary stars -- A pair of stars that regularly cross in front of each other, sometimes mimicking a planet's transit signal.

Neptunian desert -- A close-in region around stars where Neptune-sized planets are

surprisingly scarce, challenging theories of planet formation.

pipeline -- An automated chain of data-processing steps that turns raw telescope observations into a list of candidate planets.

ultra-short-period planet -- A planet that completes one full orbit around its star in less than 24 hours.

Background you need

The hunt for planets beyond our solar system began in earnest in the 1990s, and NASA's Kepler mission (2009-2018) revolutionised the field by discovering thousands of exoplanets using the transit method--measuring tiny dips in brightness when a planet crosses in front of its star. NASA's Transiting Exoplanet Survey Satellite (TESS), launched in 2018, scans almost the whole sky to find planets around closer, brighter stars.

However, not every dip is a planet. Eclipsing binary stars can create similar signals, so astronomers must carefully distinguish genuine transits from these impostors. Traditionally this was done by eye, but with millions of stars it becomes impossible. Machine learning offers a solution, and RAVEN was built to automate the validation process by training on hundreds of thousands of realistic simulations of both planets and false positives.

The 118 newly confirmed exoplanets include extreme types: ultra-short-period worlds that zip around their star in under a day, and rare Neptune-sized planets lying in the 'Neptunian desert'--a zone where such planets are unexpectedly absent. The findings sharpen our understanding of how planetary systems form, and the careful population study reduces previous uncertainties tenfold, confirming that about 9-10% of Sun-like stars have a close-in planet.

RAVEN's results are publicly available in interactive catalogues, marking a new phase where artificial intelligence not only speeds up discovery but also provides robust, statistical maps of planetary populations across the galaxy.

Break it down

METAPHOR: Finding planets among millions of stars is like searching for a needle in a cosmic haystack.

INTRODUCING RAVEN: An AI system that confirmed 118 exoplanets and flagged over 2,000 high-quality candidates from TESS data.

HOW RAVEN WORKS: Trained on hundreds of thousands of simulated planets and false signals (like eclipsing binary stars) to recognise true transits.

+ DATA SOURCE: Over 2.2 million stars observed by TESS during its first four years.

SURVEY FOCUS: Short-period planets (orbiting in less than 16 days).

+ EXTREME FINDS:

+ Ultra-short-period planets (under 24-hour orbits)

+ Worlds inside the 'Neptunian desert' that challenge theories

+ Tightly-packed multi-planet systems with newly discovered pairs.

POPULATION STUDY: Using the sample to measure how common close-in planets are - about 9-10% of Sun-like stars host one.

+ VALIDATION: Uncertainty reduced by up to a factor of ten compared to previous work.

QUOTES & RELEASE: Researchers highlight RAVEN's reliability and release interactive catalogues for global community.

Why it matters

Discoveries like these aren't just about counting planets--they help us understand how solar systems form and how many worlds might be similar to Earth. For a teenager interested in

space, AI-driven planet hunting shows that the next generation could be the one to find signs of life beyond Earth, using tools that blend astronomy, computer science and big data.

Test what you remember

1. What does the acronym RAVEN stand for?

- ☐ (A) Rapid Automated Verification of Extrasolar Novae
- ☐ (B) Rapid Automated Validation of Exoplanets from Noise
- ☐ (C) Robotic Analysis of Variable Exoplanet Networks
- ☐ (D) Remote Astrophysical Validator of Nearby Exoplanets

2. How many previously unknown exoplanets did RAVEN confirm in this study?

- ☐ (A) 18
- ☐ (B) 118
- ☐ (C) 1,000
- ☐ (D) 2,000

3. Which space telescope provided the data that RAVEN analysed?

- ☐ (A) Hubble Space Telescope
- ☐ (B) James Webb Space Telescope
- ☐ (C) Transiting Exoplanet Survey Satellite (TESS)
- ☐ (D) Kepler Space Telescope

4. What is the 'Neptunian desert'?

- ☐ (A) A dry region on the surface of Neptune
- ☐ (B) A zone around stars where few Neptune-sized planets are found
- ☐ (C) A desert on Earth named after the planet Neptune
- ☐ (D) A part of the galaxy with no stars

5. According to the companion population study, what fraction of Sun-like stars host a close-in planet?

- ☐ (A) About 1-2%
- ☐ (B) About 5-6%
- ☐ (C) About 9-10%
- ☐ (D) About 50%

6. Who led the development of the RAVEN system?

- ☐ (A) Dr. Marina Lafarga Magro
- ☐ (B) Dr. David Armstrong
- ☐ (C) Dr. Andreas Hadjigeorgiou
- ☐ (D) Dr. Kaiming Cui

Answer key (try the quiz first!): 1: B 2: B 3: C 4: B 5: C 6: C

Questions to think about

1. Positive - accelerating discovery and census: RAVEN massively speeds up the vetting of planet candidates, turning what would have taken years of manual work into a rapid, reliable process. The large sample of confirmed planets allows astronomers to make precise statements about how common different kinds of planets are, which is crucial for targeting future searches for habitable worlds.

2. Critical - limits and potential blind spots: No AI is perfect; some false positives may still be classified as planets, and the system is only as good as the simulations it was trained on. The focus on short-period planets means it says little about longer-period, potentially Earth-like planets in the habitable zone, where transits are rarer and harder to confirm.

3. Neutral - a shift in scientific methodology: This work illustrates a broader change in astronomy: from individual astronomers personally inspecting light curves to automated pipelines that handle millions of stars. It parallels the rise of big data in other fields, making science more quantitative but also more dependent on the quality of training data and algorithms.

4. **Forward-looking - paving the way for next-generation telescopes:** Future missions like ESA's PLATO telescope (due for launch in 2026) will produce even more data and demand even smarter AI to find true Earth analogues. RAVEN's approach could be adapted to search not just for transits but for atmospheric biosignatures, moving us closer to answering the question of whether we are alone in the universe.

MY THOUGHTS (at least 20 words):